Homologous Recombination

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SUSTech-BIO 2026 iGEM Team

Purpose

Assemble DNA with matching terminal overlaps using Vazyme ClonExpress Ultra One Step Cloning Kit V3 (C117). Use the rapid route for gel-purified PCR product or the calculated route for multiple fragments, followed by transformation and sequence verification.

Materials & Equipment

Item	Manufacturer	Identifier / specification	Use / Specification
ClonExpress Ultra One Step Cloning Kit V3	Vazyme	C117	2× CE Mix V3; enzyme and buffer premix [1]
Gel-purified PCR product; backbone and inserts for multi-fragment assembly	Prepared sample	M01 / M02 / M03	Correct terminal overlaps; length and concentration known for calculated assembly
DEPC-treated water	Sangon Biotech	B501005	Dilution and volume adjustment [3]
Crushed ice	Laboratory ice maker	Crushed ice bath	Reagent cooling
DNA quantification instrument	As available	DNA fluorometer / UV absorbance	Measure each DNA stock separately
PCR tubes, pipettes, filter tips; mini centrifuge	Axygen tips; Eppendorf centrifuge; laboratory equipment	Compatible sizes	Accurate reaction setup
Thermocycler	Same laboratory thermal cycler as M01	50°C incubation; 4°C hold	Controlled recombination incubation
Cloning competent cells and selection plates	As available	M05	Downstream transformation; antibiotic matched to the backbone

Before You Start

Design the homology-arm primers

Write the desired circular construct in its final order before ordering primers. For the rapid single-fragment route, the recovered PCR product must contain the sequences and terminal overlaps required to form the intended construct. For one-insert cloning, check both backbone-insert junctions; for several inserts, check every internal junction and both backbone junctions.

Use a 15-20 bp terminal overlap at each junction, preferably with 40-60% GC. A primer is written 5'-homology arm + optional retained restriction site + template-specific annealing sequence-3'. Do not count optional restriction-site or residual bases toward the recommended arm length. Calculate PCR-primer Tm from the template-specific part only; 60-65 °C is a useful design target, with the PCR program selected for the polymerase in M01. [1]

For the forward primer, take the overlap from the upstream neighbor. Write the reverse primer, including its downstream overlap, in the correct reverse-complement orientation. Simulate the PCR products and check matching ends on a common strand; a matching text string in two primer lists alone is not sufficient.

For internal junctions, place an overlap from either neighboring fragment onto the other, or split the overlap across both. Avoid repeated junction sequences that could join the wrong partners. Verify orientation, reading frame, start/stop codons and intended linkers in the assembled sequence. Count each shared overlap only once in the final construct.

Vazyme CE Design: choose C117 and the matching multi-fragment module [2]

Prepare the DNA and incubation program

Prepare PCR fragments using M01, remove susceptible parental plasmid with the separate DpnI workflow in M03 when applicable, and recover the intended bands using M02. For restriction-linearized backbones, use M03. Any digestion extension must stay within the limit of the actual enzyme; do not extend FastDigest EcoRI overnight.

Use clean eluates without visible gel debris or residual wash solution. Measure individual stocks before combining them for a calculated reaction; blank with the same eluent. Retain the sequence-derived length of each physical input fragment, including its primer-added ends.

Keep C117 at -30 to -15 °C for storage and assemble reactions on ice. Set 50 °C incubation followed by immediate cooling to 4 °C or ice. The 2× CE Mix V3 already supplies the reaction enzymes and buffer. [1]

Procedure

A. Rapid single-fragment recombination

1. Take 10 µL of PCR product purified by agarose gel electrophoresis and gel extraction (M02). Use the recovered DNA eluate for the reaction.

2. On ice, add 5 µL of 2× CE Mix V3 to the 10 µL DNA eluate, using the following 15 µL reaction.

Component	Volume
Gel-purified PCR product	10 µL
2× CE Mix V3 (C117)	5 µL
Total	15 µL

3. Gently pipette to mix, then briefly spin to collect the liquid at the bottom of the tube.

4. Incubate at 50°C for 5 min, then immediately cool to 4°C or place on ice. Proceed to the controls and downstream handoff in section C.

This 10 + 5 µL setup is the laboratory-adapted rapid recipe. Its final CE Mix concentration is approximately 0.67×; the manufacturer reference uses 5 µL of 2× Mix in 10 µL total (1×). The rapid recipe is retained here as used; the calculated multi-fragment route below follows the 1× reference setup. [1]

B. Multi-fragment recombination: calculate each input

5. List the backbone and every insert in assembly order with length L (bp) and measured concentration C (ng/µL). Here, n is the number of inserts: C117 supports up to 10 inserts plus one backbone, or 11 DNA pieces in total. [1]

For double-stranded DNA of length L, approximate molecular weight MW = 660 × L g/mol. Convert mass to molar amount q using q (pmol) = mass (ng) / (0.66 × L). Conversely, mass (ng) = 0.66 × L × q, and volume (µL) = mass / C. Equal masses of different-length fragments contain different numbers of molecules.

6. Start with approximately 0.03 pmol of each DNA piece, giving a backbone:each insert molar ratio of 1:1. The C117 shortcut is mass (ng) ≈ 0.02 × L (bp). If this falls outside 10-100 ng, use the corresponding 10 ng minimum or 100 ng maximum recommended by C117, then calculate and record the resulting pmol. Mass limits can make the final ratio depart from exact equimolarity. [1]

Worked example only: a 4,000 bp backbone plus three inserts. The rounded 0.02 × L rule gives approximately 0.0303 pmol of each physical DNA piece.

DNA piece	Length (bp)	C (ng/µL)	Mass (ng)	Volume (µL)
Backbone	4,000	80	80	1.0
Insert A	1,000	20	20	1.0
Insert B	1,500	30	30	1.0
Insert C	500	10	10	1.0

7. Sum all DNA volumes before adding Mix. For the 10 µL reference reaction, DNA plus water must total 5 µL. In the example, DNA totals 4 µL, leaving 1 µL water. Dilute an individual stock if a required addition is too small to pipette accurately, then recalculate the volume budget.

Component	10 µL reference reaction	Worked example
Linearized backbone	Vbackbone µL	1 µL
All inserts	V1 + V2 + ... + Vn µL	3 µL combined
2× CE Mix V3	5 µL	5 µL
Nuclease-free water	5 - total DNA volume (µL)	1 µL
Total	10 µL	10 µL

For this calculated 1× route, if DNA volumes exceed 5 µL, concentrate the stocks or lower inputs within C117's compatible 0.003-0.25 pmol range; keep the recommended mass range in view. For many pieces, a larger 1× reaction may be needed for accurate additions. Set Mix to half of the chosen total volume and document the scale adaptation. [1]

8. Combine the calculated DNA and water on ice, add 2× CE Mix V3, gently pipette and briefly spin. Incubate at 50°C for 15-30 min. Use 30 min when assembling more than five inserts; immediately cool to 4°C or ice. Do not extend beyond the C117 recommended window as a routine rescue. [1]

C. Controls and downstream handoff

9. For the rapid route, prepare a control containing 10 µL of the same gel-purified PCR product and 5 µL water, without CE Mix. For multi-fragment assembly, prepare a backbone-only no-Mix control and, when an insert template carries the same resistance, an insert-only no-Mix control. Match DNA input and final volume to the test. These controls help detect residual circular plasmid; a kit positive control helps assess reagents and transformation. [1]

10. Proceed to M05 for transformation using cloning competent cells and the backbone's selection marker. For chemical transformation, keep assembly-product volume at or below one-tenth of competent-cell volume. C117 products may be stored at -20°C for up to one week if transformation is deferred. [1]

11. Screen individual colonies across the designed junctions, then sequence the junctions and PCR-derived regions needed to confirm the construct. Record the DNA stocks, lengths, masses, volumes, incubation and clone identities so that a successful assembly can be reproduced.

p.s.

A fixed eluate volume does not fix DNA mass. Check concentrations when a new DNA preparation replaces the previous stock.

Review every internal overlap before repeating a multi-insert assembly; more enzyme cannot repair a missing or incorrectly oriented junction.

For gel-observed defects, use M02 first. Refer amplification specificity to M01 PCR Optimization O2 and parental-template removal to the separate M03 DpnI workflow.

Expected Outcome

The cooled reaction is ready for transformation. Successful clones should give the predicted junction-spanning PCR products and sequence agreement with the designed construct. Neither reaction appearance nor colony growth alone confirms complete, correctly ordered assembly; interpret test colonies alongside the controls.

Troubleshooting

Resolve shared transformation or selection problems in M05 first when controls fail together. Use the cases below for assembly-specific follow-up.

Observed problem	Check first	Action
Few colonies; positive control grows	DNA amounts, terminal overlaps, Mix dilution and cleanup	Recalculate DNA inputs and inspect every junction. If the rapid route gives poor recovery, compare with the manufacturer's 1× setup using matched DNA amounts.
Colonies in a no-Mix DNA control	Residual circular backbone or insert template	Use the separate controls to localize carryover; reprepare the affected DNA via M03 and M02. Respect enzyme-specific limits.
Colonies lack an insert or contain the wrong order	All junction sequences and the identity of input stocks	Check orientation and repeated overlaps in silico; screen across internal junctions and sequence candidates.
Colony PCR gives no expected product	PCR controls and whether primers flank an informative junction	Use at least one backbone primer where appropriate; test extracted plasmid or an informative digest before declaring assembly failure.
Calculated DNA volumes do not fit	Units, stock concentrations and total DNA volume	Concentrate, adjust inputs within C117 guidance, or use a documented larger 1× setup. Recalculate water; do not omit Mix.

Reference

[1] Vazyme. ClonExpress Ultra One Step Cloning Kit V3 (C117), Instruction for Use, Version 24.1. Sections 06-3, 07-1 to 07-3 and 08; printed pp. 4-10. Manufacturer manual hosted by Biorbyt; downloaded and checked 2026-09-16. Open source

[2] Vazyme. CE Design, Multi-Fragment Assembly module; includes C117 product selection and vector-linearization inputs. Accessed 2026-09-16. Open source

[3] Sangon Biotech. B501005 DEPC-treated water, SDS, revision 2024-12-24; product identity. Accessed 2026-10-03. Open source